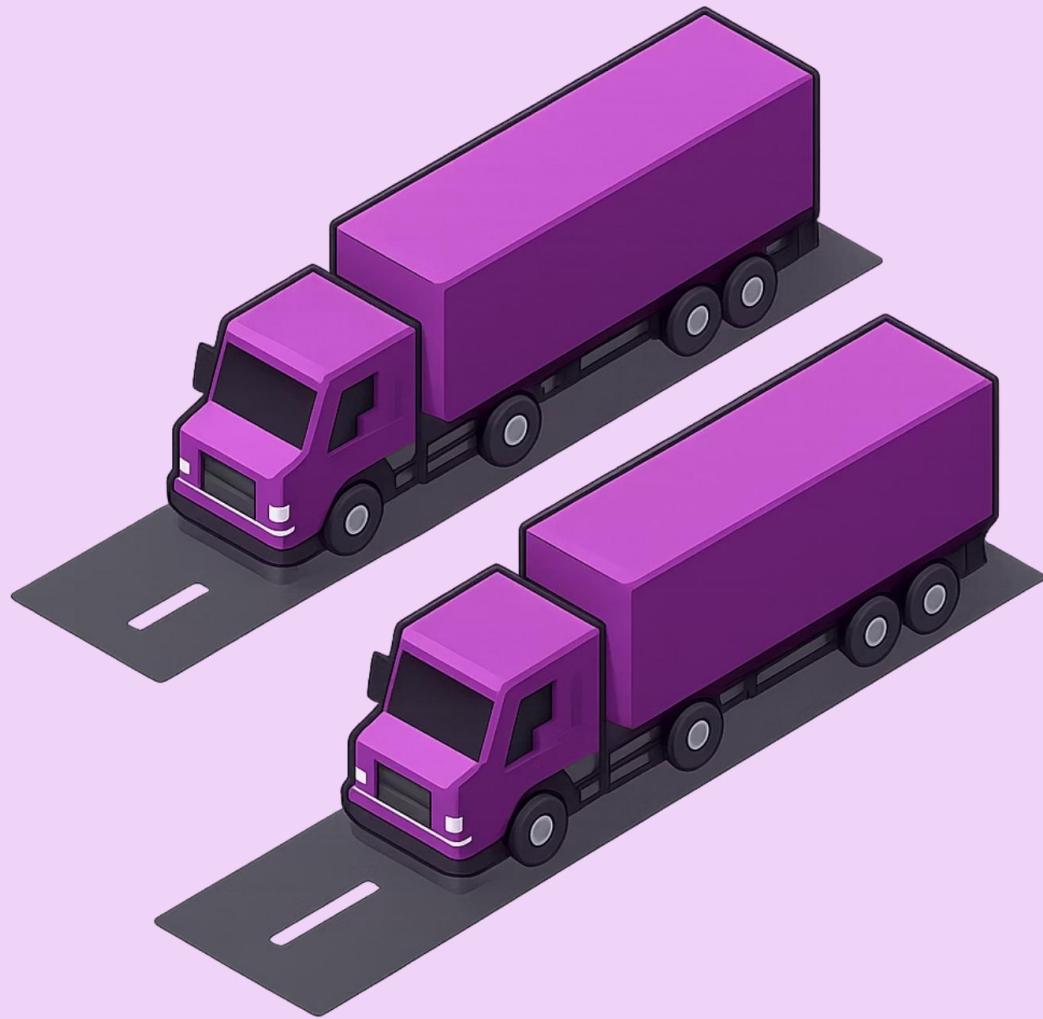




Moving What Matters

Cargo-Core: Revolutionizing Rural Logistics

A high-tech platform built for the unique demands of small business operations in rural India — bridging the gap between traditional logistics and digital transformation.

Agenda

What we'll cover today — from the problem we're solving to the technology powering our solution.

01

Introduction & Team

Meet QuadCore-Devs

02

Vision & Users

Who we serve and why

03

User Challenges & Needs

Pain points in rural logistics and their needs

04

Platform Architecture

How Cargo-Core works

05

Smart Operations

Innovations that set us apart

06

Automation & Financials

Efficiency at scale

07

Tech Stack & Deliverables

Modern Technologies for Rural Challenges

08

Conclusion & Vision

The road ahead



CHAPTER 1

Introduction & Team

Who we are, what we've built, and why we're the right team to solve this problem.

Welcome to Cargo-Core

Cargo-Core is a next-generation logistics platform purpose-built for rural India — unifying booking, dispatch, inventory, and payments in one intelligent system.



Rural-First Design

Built for the unique needs of rural India, bridging traditional operations with digital transformation.



Unified Operations

Streamlines booking, dispatch, inventory, and payments — reducing manual errors and boosting efficiency.



Smart & Scalable

Modern dashboards, real-time tracking, and smart automation — architected for AI-ready growth.

Meet the Quad-Core-Devs Team

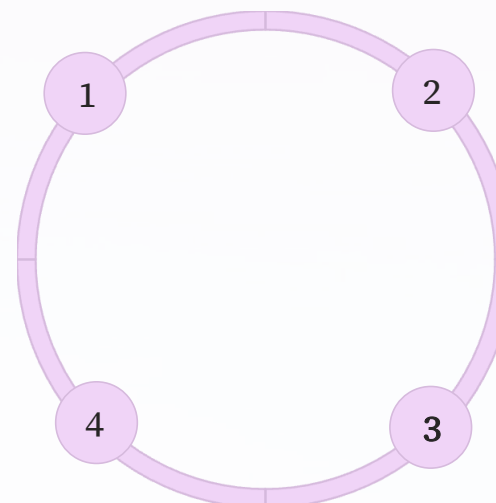


Team-003 · Indian Institute of Technology, Madras

A cross-functional team driven by agile methodology, collaboration, and high-impact innovation. Each member brings a distinct skill set — from product strategy to full-stack development and quality assurance.

Siddarth S
Project Manager & Scrum Master

Gautham Krishna S
Backend Developer & Tester



Yashvardhan
Full-Stack Developer & Code Reviewer

Pruthvi Prasad S
Backend Developer & Frontend Developer

An isometric illustration of a road network in shades of purple and black. The roads are represented by parallel lines forming a grid. Several rectangular areas, possibly representing fields or construction sites, are interspersed within the grid. Two small, stylized trucks are shown on the roads: one in the lower-left corner and another further up the network. The background is a solid light purple color.

CHAPTER 2

Vision & Users

Our mission is to bring digital infrastructure to the backbone of India's rural economy — and we've designed Cargo-Core for every stakeholder in the logistics chain.

Vision for Rural Logistics

Cargo-Core replaces fragmented, manual coordination with a unified digital platform — giving every stakeholder the visibility and tools they need to operate with confidence.



From Manual to Digital

Replacing phone-based coordination with structured, trackable workflows that reduce errors and confusion.



Accountability & Trust

Automated records for payments, deliveries, and resource allocation — eliminating disputes and building trust.



Real-Time Visibility

Centralized dashboards for live tracking of orders, vehicles, and inventory across the entire operation.



Scalable Growth

Structured workflows designed to support expansion and handle peak-season demand without breaking down.

Who Uses Cargo-Core?

Primary User

01 · Operations Core

- *Warehouse Manager* — oversees inventory, storage, and dispatch readiness
- *Dispatcher* — coordinates vehicle and labor assignments per order
- *Driver* — executes deliveries and manages on-ground logistics

Secondary User

02 · Strategic & Service

- *Owner / Logistics Manager* — monitors bookings, payments, and overall operations
- *Customer* (Farmer, Vendor, Individual) — books services and tracks deliveries

03 · Indirect Workforce

- *Laborers / Helpers* — managed by primary users for loading and unloading tasks

Tertiary User





CHAPTER 3

User Challenges & Needs

Before building Cargo-Core, we listened. These are the real operational failures costing rural logistics businesses time, money, and trust every single day.

Understanding User Pain Points

Manual Operations Create Bottlenecks

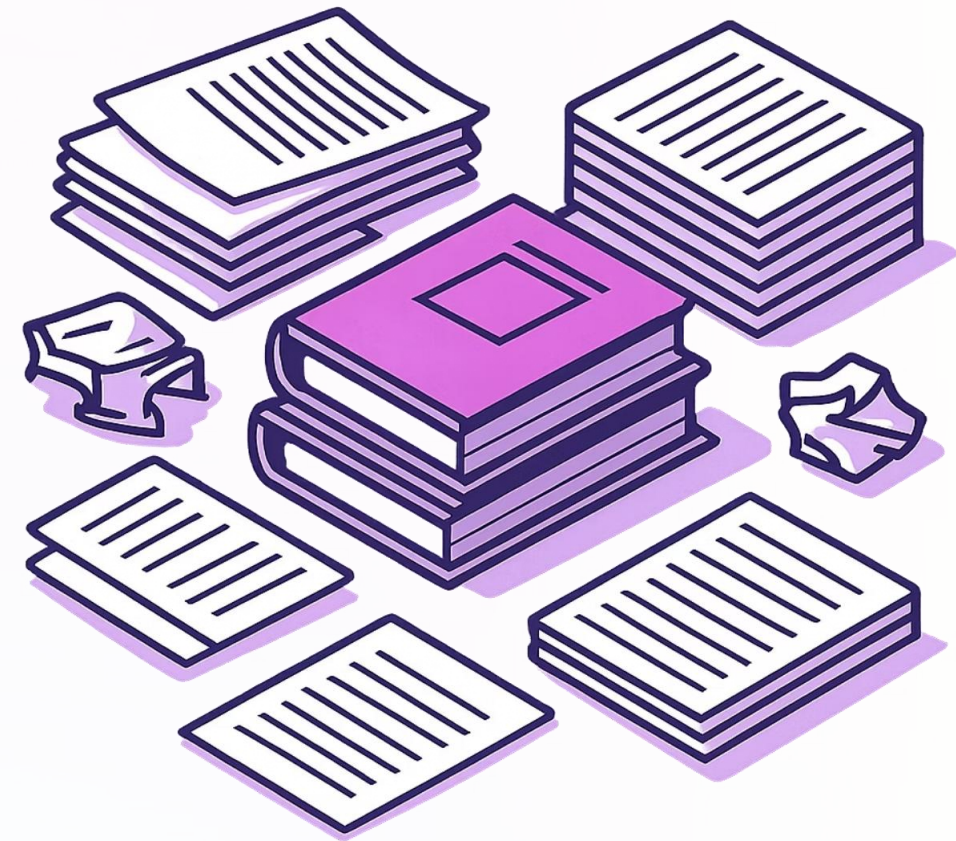
Phone calls and paper records cause confusion and missed details. The owner becomes a single point of failure, creating operational overload.

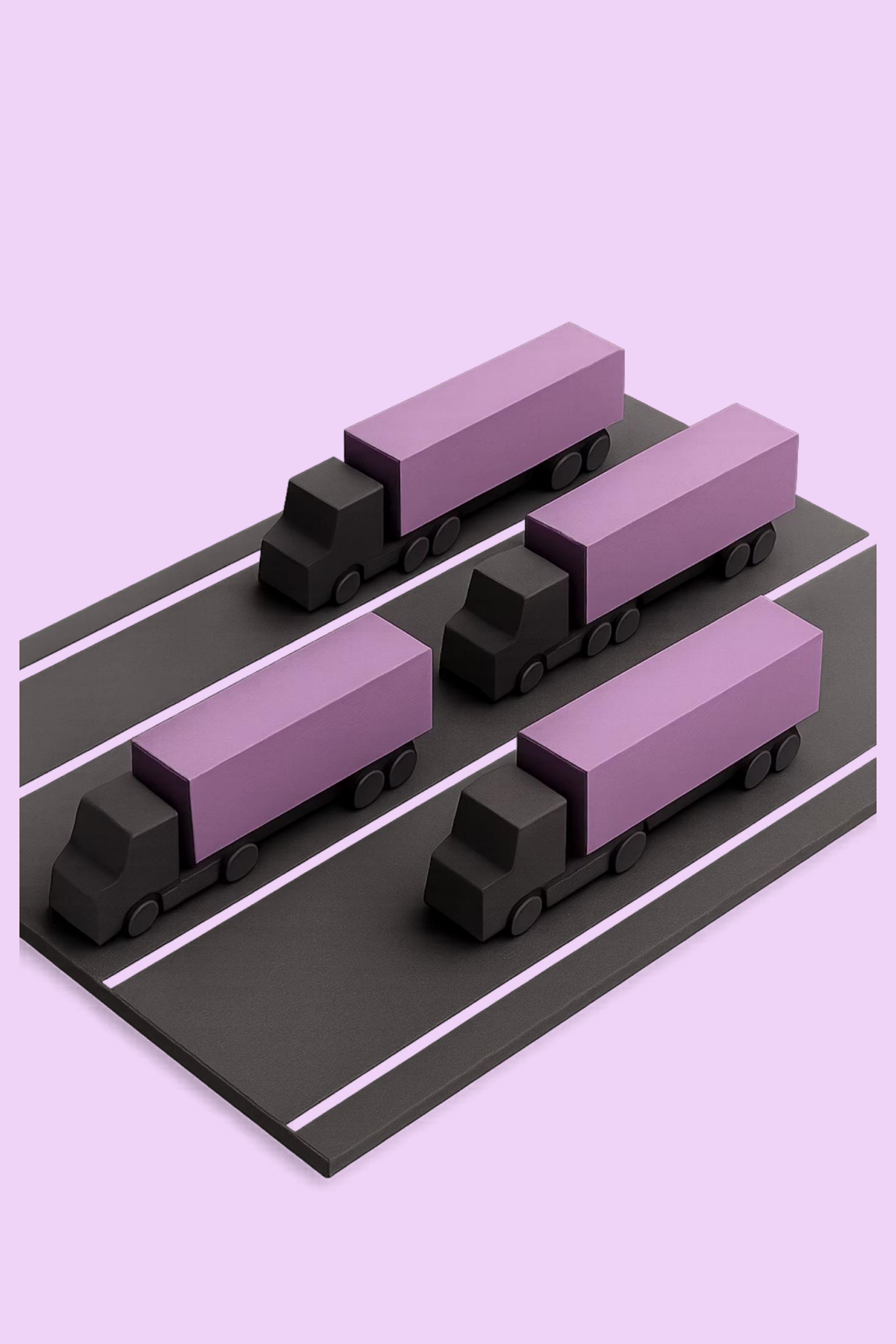
No Real-Time Visibility

Without live tracking of orders, vehicles, or payments, stakeholders operate in the dark — leading to delays and disputes.

Resource & Financial Inefficiencies

Guesswork drives vehicle and labor misallocation. Manual billing and payroll result in pending dues and suspected fuel fraud.





USER CHALLENGES & PAIN POINTS

Where Logistics Operations Break Down

From manual invoices to missing delivery visibility, small-to-midsize logistics operators face systemic inefficiencies that erode profitability and customer trust — especially during peak demand.

OWNER CHALLENGES

Owner's Daily Struggles

Financial Bottlenecks

Pending customer dues strain cash flow. Manual invoice generation introduces errors and delays collections.

Lack of Real-Time Oversight

No live vehicle tracking creates delivery uncertainty. Manual payroll calculations cause costly delays.

Peak Season Overload

Operational stress spikes during high-demand periods. Resource allocation and scheduling become overwhelming.



Customer Experience Gaps



No Booking Confirmation

Customers receive no digital order summary, leading to miscommunication and disputes.



Unclear Goods Specification

Phone-based descriptions cause resource underestimation and operational delays.



No Delivery Tracking

Without live status visibility, delays directly impact customer income and satisfaction.

Warehouse Management Issues

Chaotic Storage

Unmapped sections lead to misplaced goods and time-consuming searches.

Equipment Loss

No digital tracking for crates, belts, and returnables means frequent losses.

Packing Shortages

Absence of inventory alerts causes last-minute shortages and dispatch delays.

No Inventory Tracking

Manual records prevent real-time visibility and efficient stock management.



Dispatcher's Daily Struggles

1

Manual Assignments

Resource allocation relies on gut judgment, causing miscalculations and unsuitable vehicle-labor pairings.

2

Unstructured Scheduling

Absence of a digital schedule creates overlaps, missed deadlines, and no visibility into job progress.

3

No Decision Support

Without live data or contingency tools, dispatchers can't proactively respond to breakdowns or peak loads.



Field & Workforce Gaps

Driver Challenges

- ***Incomplete job info*** — No digital manifests increase missed or delayed deliveries
- ***No route optimization*** — Longer travel times and higher fuel costs
- ***No proof of delivery*** — Causes disputes and payroll inaccuracies
- ***Connectivity gaps*** — Rural dead zones disrupt coordination

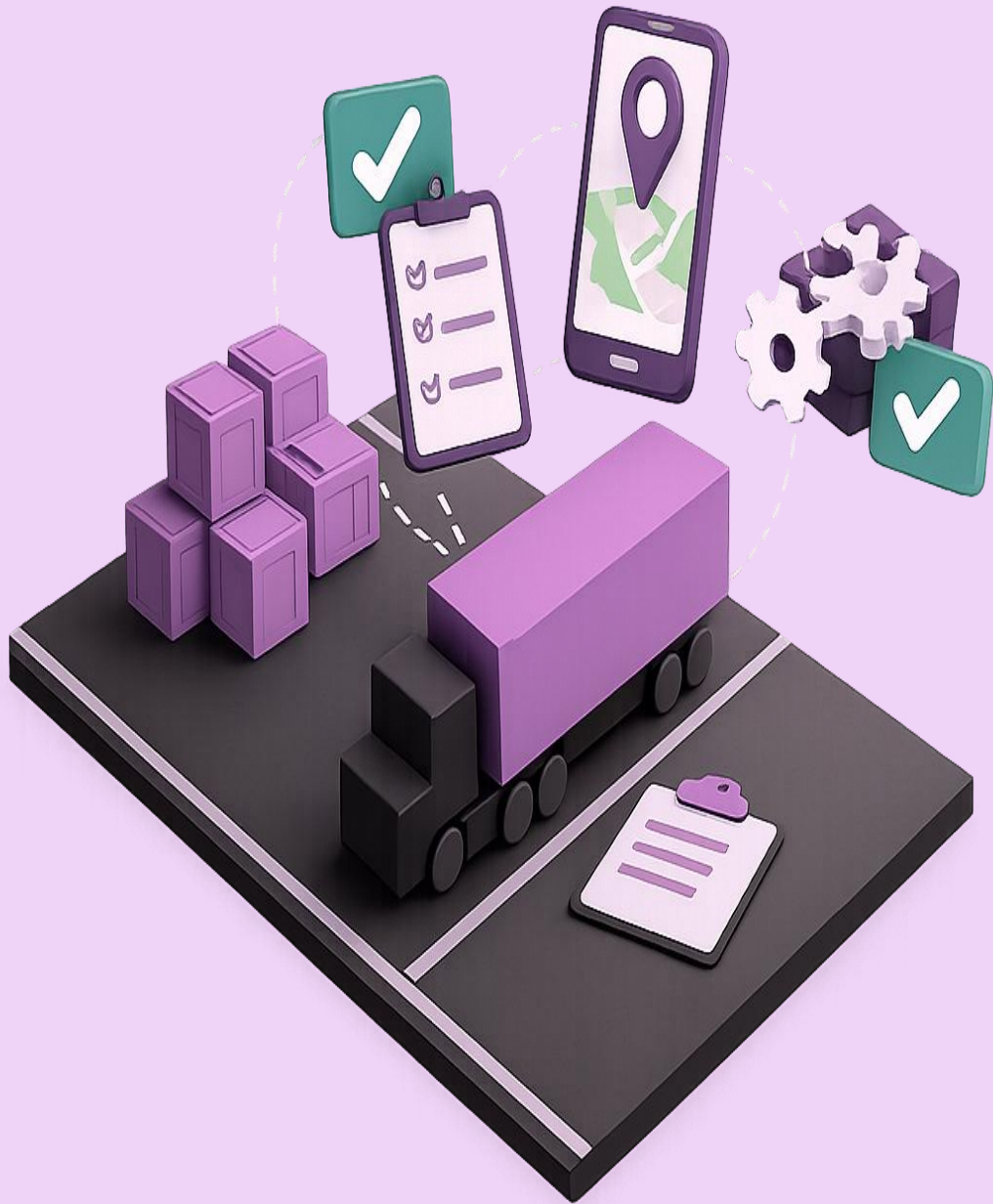
Labour Management Gaps

- ***No structured scheduling*** — Attendance and shifts tracked informally
- ***Informal payroll*** — Payments managed without digital records
- ***Resource underestimation*** — Labour needs misjudged, causing delays

The Three Root Causes of Inefficiency



These three systemic gaps — disconnected booking, fragmented data, and absent automation — compound across every role, from owner to driver, creating a cycle of manual work, errors, and lost revenue.



USER CHALLENGES & NEEDS

Where User Challenges Arise

Manual processes, lack of tracking, and poor coordination create operational breakdowns across booking, dispatch, and delivery — directly impacting efficiency and user trust

User Challenges & Needs

*Eight core epics drive the platform's design — from booking and billing to fleet management, workforce coordination, and asset protection. Each story follows the **S.M.A.R.T. framework**: Specific, Measurable, Achievable, Relevant, and Time-bound.*

Order Booking

Structured requests, confirmations, and resource suggestions

Equipment Protection

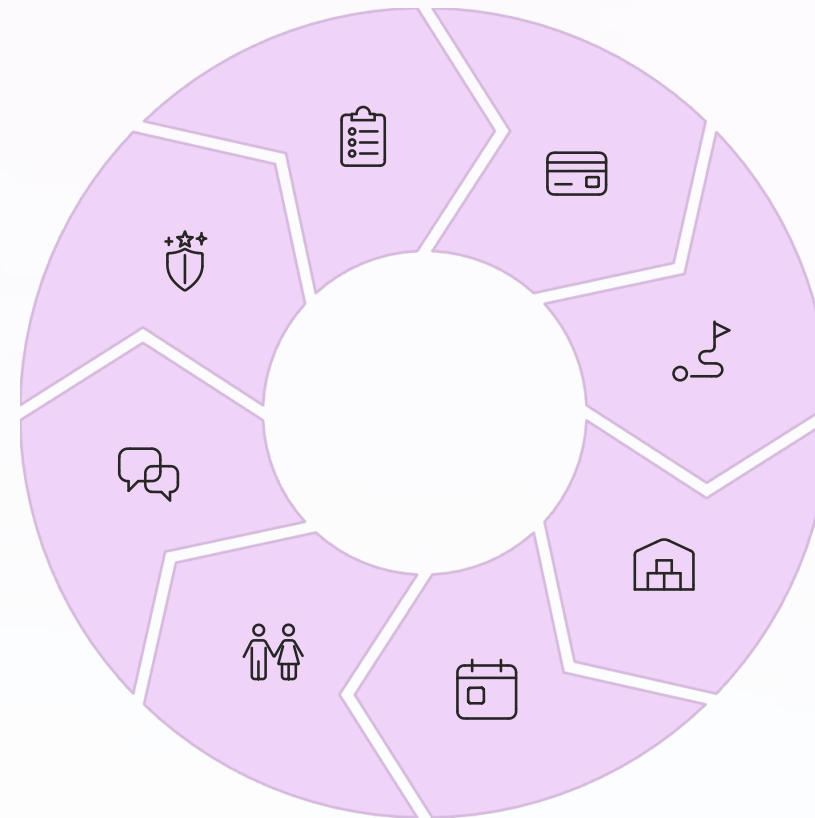
Accountability logs and fuel usage monitoring

Communication

Centralized comms, offline sync, and peak-season prioritization

Workforce & Payroll

Digital hours, attendance, and weekly payroll summaries



Payment & Billing

Auto-invoicing, payment tracking, and partial balance clarity

Fleet & Route

Optimized routing, live tracking, and status updates

Warehouse & Inventory

Mapped storage, low-stock alerts, and equipment tracking

Resource Scheduling

Automated assignment, daily schedules, load time estimates

User Stories by Epic

S.M.A.R.T. FORMAT: AS A [USER], I WANT [ACTION], SO THAT [BENEFIT]

1

Epic 1 — Order Booking & Confirmation

- **1.1 Structured Request:** As a Customer, I want to submit a structured booking request with goods, quantity, and location details — so the provider understands requirements without confusion.
- **1.2 Confirmation Receipt:** As a Customer, I want a digital confirmation after booking — so I have written proof of order details and cost.
- **1.3 Cost & Resource Suggestion:** As a Customer, I want the system to suggest vehicle size and labour count — so I avoid underestimating resources and prevent delays.

2

Epic 2 — Payment & Billing Management

- **2.1 Auto Invoice:** As a Owner, I want the system to auto-generate invoices for every completed order — so manual receipt creation is eliminated.
- **2.2 Payment Tracking:** As a Owner, I want to track full, partial, and pending payments — so outstanding dues are visible and revenue loss is reduced.
- **2.3 Flexible Recording:** As a Customer, I want to record partial payments and remaining balances clearly — so both parties maintain transaction transparency.

3

Epic 3 — Fleet & Route Management

- **3.1 Route Optimization:** As a Driver, I want optimized navigation routes — so I reduce fuel consumption and avoid unnecessary backtracking.
- **3.2 Live Tracking:** As a Owner, I want real-time vehicle location visibility — so I can monitor deliveries without repeatedly calling drivers.
- **3.3 Status Updates:** As a Customer, I want delivery status updates pushed to me — so I don't need to call for information.

4

Epic 4 — Warehouse & Inventory Management

- **4.1 Mapped Storage:** As a Warehouse Manager, I want goods stored with mapped sections and labeled locations — so items can be found quickly at dispatch.
- **4.2 Low-Stock Alerts:** As a Warehouse Manager, I want automatic alerts when materials run low — so restocking happens before delays occur.
- **4.3 Equipment Tracking:** As a Warehouse Manager, I want issued and returned equipment recorded per order — so losses of crates, belts, and blankets are minimized.

User Stories by Epic (continued)

S.M.A.R.T. FORMAT: AS A [USER], I WANT [ACTION], SO THAT [BENEFIT]

1

Epic 5 — Resource Allocation & Scheduling

- **5.1 Automated Assignment:** As a Manager, I want automated job assignment based on availability and skills — so dispatching is efficient and fair.
- **5.2 Daily Schedule:** As a Worker, I want a clear, daily schedule on my app — so I know what jobs are assigned to me and when.
- **5.3 Load Time Estimation:** As a Dispatcher, I want load time estimates for each job — so I can plan routes efficiently and avoid idle time.

2

Epic 6 — Workforce & Payroll Management

- **6.1 Digital Hours:** As a Worker, I want to digitally log my start and end hours for each job — so my pay is accurately calculated without manual timecards.
- **6.2 Attendance Tracking:** As a Manager, I want to track worker attendance and punctuality — so I can identify potential issues and ensure timely job completion.
- **6.3 Weekly Payroll Summary:** As an Owner, I want a weekly payroll summary generated automatically — so I can quickly approve payments and reconcile accounts.

3

Epic 7 — Communication & Coordination

- **7.1 Centralized Comms:** As a Manager, I want a centralized communication channel for all job-related messages — so critical updates aren't missed across multiple platforms.
- **7.2 Offline Sync:** As a Worker, I want the app to sync messages and job details offline — so I can access information even without an internet connection in remote areas.
- **7.3 Peak-Season Prioritization:** As a Dispatcher, I want the ability to prioritize urgent messages during peak seasons — so critical information gets immediate attention.

4

Epic 8 — Equipment & Asset Protection

- **8.1 Accountability Logs:** As a Manager, I want a clear log of who is assigned which equipment — so accountability for assets is maintained.
- **8.2 Fuel Usage Monitoring:** As an Owner, I want to monitor fuel usage per vehicle and per job — so I can identify inefficiencies and reduce operational costs.

Why These Stories Matter

Each user story directly addresses a pain point, ensuring the platform delivers tangible value by improving operational efficiency, reducing costs, and enhancing user experience across all roles within the organization.

CHAPTER 4

Platform Architecture

A Cargo-Core utilizes a secure, unified cloud architecture designed to seamlessly connect all rural logistics stakeholders on a single platform. A modular backend engine orchestrates real-time data flow between a centralized database and dedicated web and mobile interfaces for drivers, owners, and dispatchers.



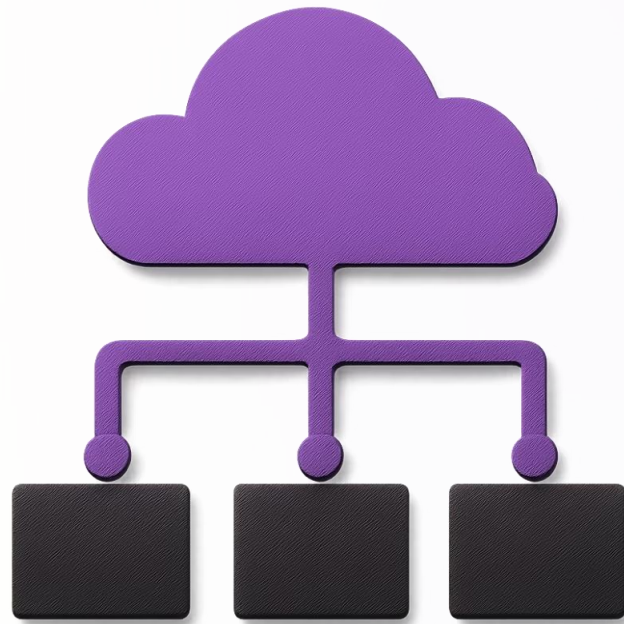


PLATFORM ARCHITECTURE

Introducing Cargo-Core

A unified, cloud-based logistics platform built to eliminate manual bottlenecks — connecting owners, warehouses, dispatchers, drivers, and customers on a single source of truth.

Cargo-Core: High-Level Architecture



Unified Cloud Platform

All roles share a single centralized database for real-time data consistency.

Modular Backend Engine

Handles routing, inventory, payments, and AI-powered optimization at scale.

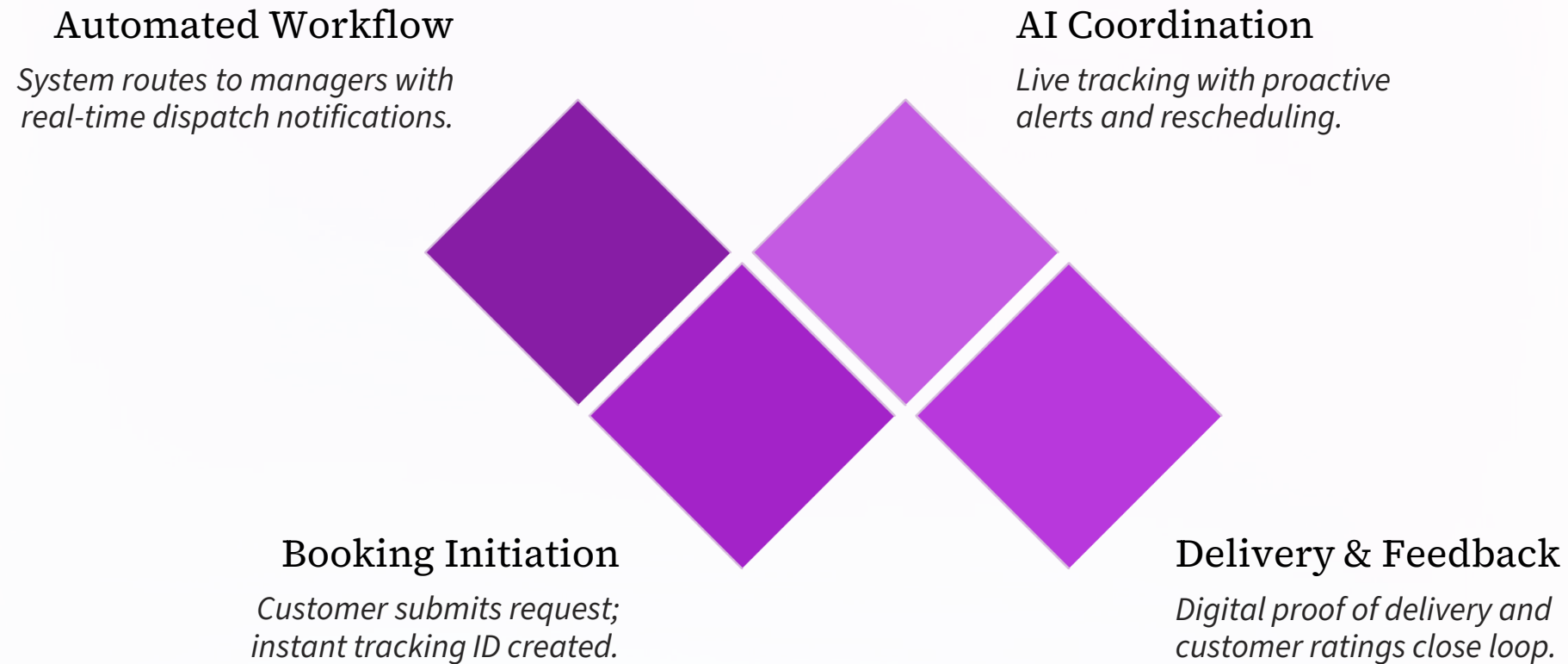
Role-Based Dashboards

Six tailored interfaces — Owner, Warehouse, Dispatcher, Driver, Customer, Support Bot.

Seamless Workflow Integration

Automated hand-offs ensure smooth order lifecycle from booking to delivery.

Order Lifecycle: End-to-End Flow



Cargo-Core automates every hand-off — from instant booking confirmation through AI-driven coordination to digital proof of delivery — giving every stakeholder real-time visibility and closing the service loop with customer feedback.

PLATFORM ARCHITECTURE

Role-Based Dashboards & Platform Workflow

Every stakeholder — from warehouse managers to drivers — accesses a tailored interface with only the data and controls relevant to their role, ensuring clarity, security, and operational efficiency across the entire logistics chain.





PLATFORM ARCHITECTURE

Real-Time Intelligence, Security & Analytics

Live Operational Maps

Vehicle locations, delivery routes, inventory, and order status updated in real time.

Role-Based Access Control

Granular permissions protect data privacy. Audit logs track all high-level actions.

Actionable Analytics

Dashboards visualize KPIs and bottlenecks. AI recommends resource optimizations.



CHAPTER 5

Smart Operation & Innovation

Cargo-Core leverages AI-driven analytics and automated resource allocation to proactively optimize routing, fleet management, and crew assignments in real time. Our intelligent workflows provide live inventory visualization and smart fulfillment tracking to eliminate manual errors and equipment losses. By integrating generative AI assistance and continuous IoT monitoring, the platform empowers rural logistics teams to operate with the efficiency of modern enterprise supply chains.

Smart Resource & Fleet Management



Automated Resource Assignment

System recommends optimal vehicles and laborers per order. Dynamic crew manifests ensure the right team and equipment for every job.

Optimized Route Planning

Real-time optimization using traffic and vehicle constraints for fuel efficiency. Live GPS tracking provides full delivery transparency.

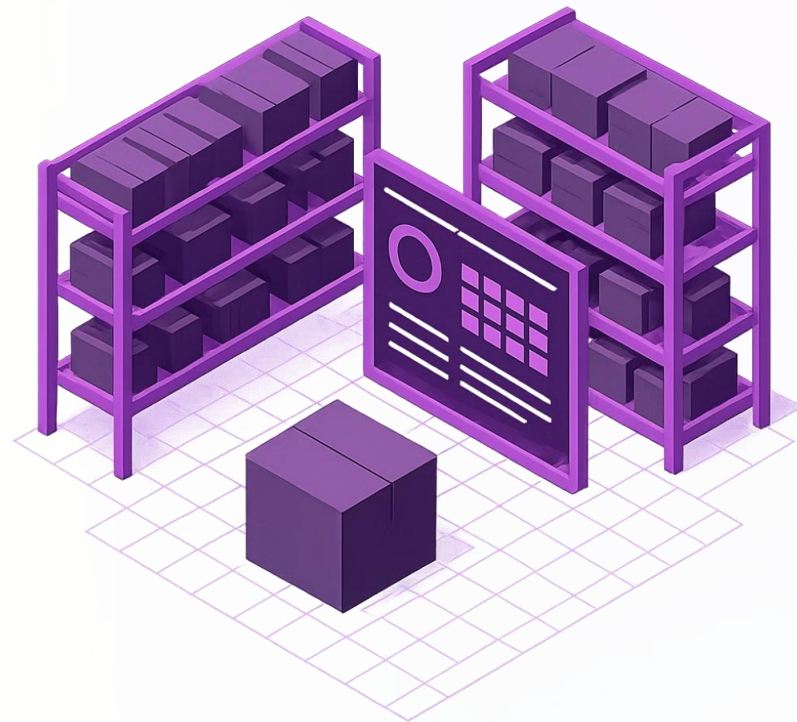
Geofencing & Crisis Management

Geofenced zones trigger auto-arrival and security alerts. Instant re-routing and emergency tools enable rapid disruption response.

Performance Metrics

Monitor on-time delivery, empty miles, and route deviation. AI-driven insights power continuous improvement.

Warehouse & Inventory Innovations



Live Inventory Visualization

Track every SKU in real time with mapped storage locations. Digital floor plans highlight available and occupied bins, streamlining warehouse navigation.

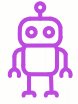
Smart Fulfillment & Packing

Intelligent pick-lists optimize walking paths and reduce double-handling. Packing station interfaces verify items and auto-generate shipping labels.

Asset & Returns Management

Returnable equipment tracked from issue to return, minimizing losses. Reverse logistics workflows handle inspections, restocking, and damage claims efficiently.

AI-Powered Customer Support



Always-On Smart Support

AI-powered bot delivers instant answers, order status updates, and handles booking changes — day or night — without human delay.



Real-Time Tracking & Proof

Integrates live GPS for accurate driver location and ETA. Shares delivery photos and packing confirmation instantly.

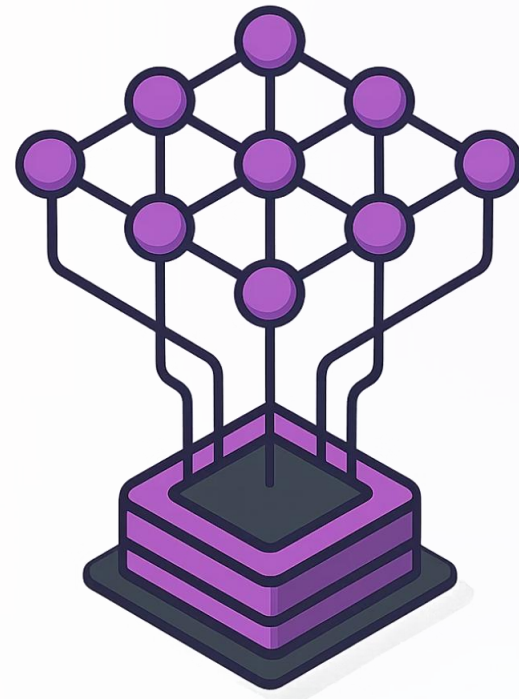


Seamless Human Escalation

Detects complex issues and routes them to a human manager automatically, ensuring a smooth handoff for unresolved problems.



Generative AI Across the Platform



AI-Driven Analytics

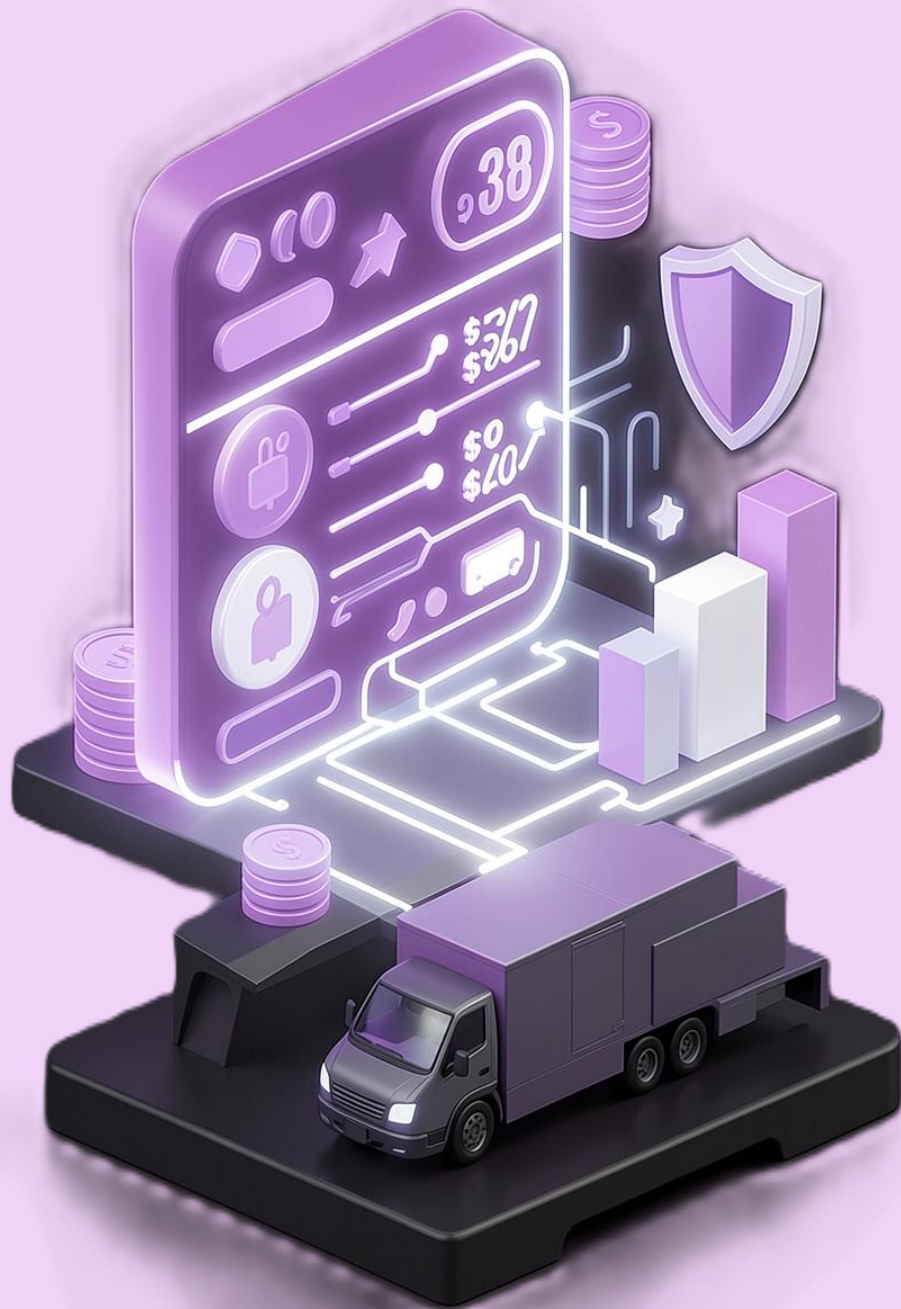
Natural language queries deliver instant data-backed summaries. Predictive optimization reallocates staff based on live demand.

Automated Detection & Reporting

AI scans transactions and GPS logs to flag fraud or anomalies. Daily operational health reports generated automatically.

Smart Estimation & Assistance

Visual volume estimator recommends packing materials from photos. Voice-activated driver assistant provides hands-free route updates.



CHAPTER 6

Automation & Financials

Cargo-Core eliminates manual financial bottlenecks by automatically generating digital invoices and instant receipts the moment a delivery is completed. The platform features a centralized dashboard for real-time tracking of full, partial, and pending payments, providing clear oversight to reduce revenue leakage. Additionally, it automates payroll based on recorded work hours and utilizes smart expense auditing to compare fuel usage against actual mileage, instantly flagging any anomalies.

Financial & Payroll Automation



Seamless Digital Invoicing

Automatic invoice generation for every completed order. Instant digital receipts sent to customers for full transparency.



Real-Time Payment Tracking

Monitor full, partial, and pending payments in a unified dashboard. Clear outstanding dues overview reduces revenue leakage.



Automated Payroll & Auditing

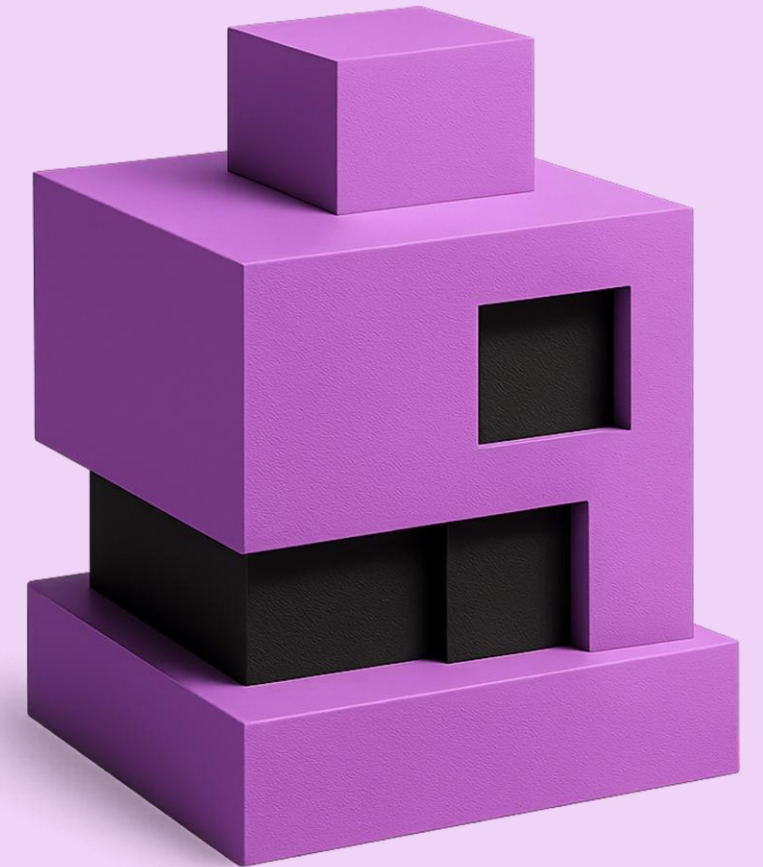
Payroll calculated from recorded work hours and performance metrics. Expense auditing compares fuel usage with mileage to flag anomalies.



CHAPTER 7

Tech Stack & Deliverables

A deep dive into the technologies powering Cargo-Core — and the milestones that brought it to life.



Built on a Purpose-Driven Stack

Every technology in Cargo-Core was chosen for performance, scalability, and real-world fit in rural logistics.



Backend & Core Logic

Python, FastAPI, and Uvicorn power business logic, APIs, and async server performance.



Driver App (Mobile)

Vue 3 + Vite, Capacitor 6, and Tailwind CSS 3 support the mobile driver app.



Mapping & Geolocation

Leaflet and Vue-Leaflet enable interactive maps, vehicle tracking, and route visualization.



Frontend (Web App)

Vue 3, Vite 6, Pinia, Vue Router 4, and Tailwind CSS 4 deliver the web experience.



Database Architecture

PostgreSQL, asyncpg, SQLAlchemy, and Alembic manage secure data storage and migrations.



Artificial Intelligence

Gemini API and google-genai enable forecasting, smart assignment, and automated support.

Frontend, Mobile & Infrastructure

Advanced UI, mobile capabilities, and infrastructure tools powering the platform.



Charts & Visualization

***Chart.js** and **Vue-Chartjs** enable rich data visualization, interactive dashboards, and real-time analytics across the platform.*



3D & Advanced Features

***Spline Viewer** brings 3D visualization capabilities, while **qrcode.vue** enables quick QR code generation for tracking and sharing.*



Authentication & Security

***JWT**, **python-Jose**, **bcrypt**, and **Google OAuth** provide secure authentication, encrypted sessions, and trusted user sign-in.*



Animations & UI

***GSAP**, **Vue-Use Motion**, **Lucide Vue**, and **Material Symbols** create smooth animations, polished interactions, and consistent visual design.*



Mobile Native Features

***Camera**, **Geolocation**, **Speech Recognition**, **Text-to-Speech**, **Barcode Scanning**, and **Push Notifications** unlock native device capabilities for the driver app.*

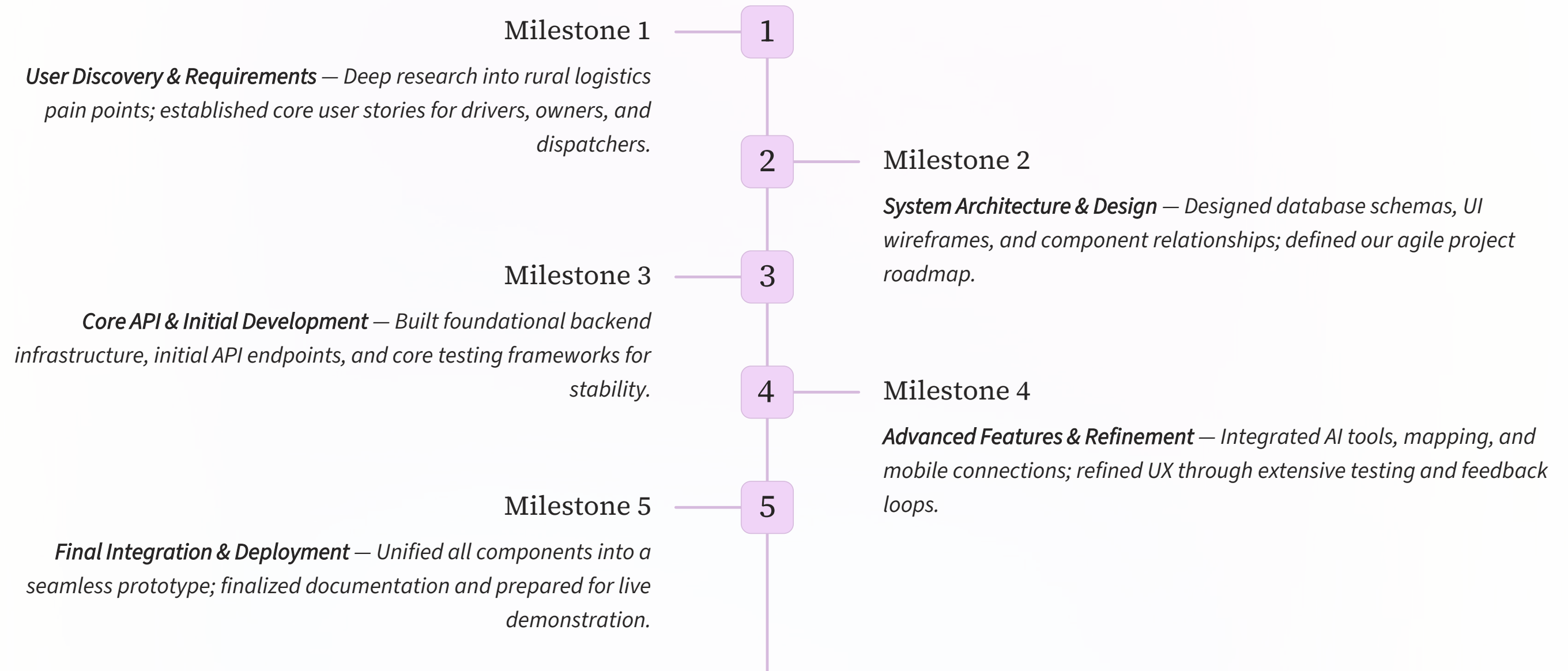


Background Tasks & Logging

***Celery**, **Loguru**, **Redis**, **pytest**, and **httpx** handle async job processing, comprehensive logging, caching, and reliable testing.*

Development Milestones

From discovery to deployment — five structured phases that shaped Cargo-Core into a production-ready platform.



Project Deliverables

All artifacts from the Cargo-Core project are organized and accessible below. Scan the QR code or use the individual links to explore each resource.

Demo Video

Watch the full live walkthrough of the Cargo-Core platform in action. Link to be provided.

Project Reports

Access all milestone reports and technical documentation. Link to be provided.

GitHub Repository

Browse the full source code, commit history, and CI/CD configuration. Link to be provided.

GitHub Wiki

Developer documentation, setup guides, and architecture notes. Link to be provided.

Project Website

Visit the live Cargo-Core landing page and product overview. Link to be provided.

Scan to Access

One QR code — your gateway to all Cargo-Core deliverables. Scan with any mobile device to get instant access to the demo, repo, and documentation.





CHAPTER 8

Conclusion & Vision

A unified, cloud-based logistics platform built to eliminate manual bottlenecks — connecting owners, warehouses, dispatchers, drivers, and customers on a single source of truth.

Transforming Rural Logistics



From Manual to Digital

Replaces paper-based chaos with a unified, cloud-powered platform. Centralizes all logistics data for instant access and control.



Empowering Every Role

Custom dashboards for Owner, Warehouse, Dispatcher, Driver, and Customer. Real-time tracking, automated alerts, and transparent workflows.

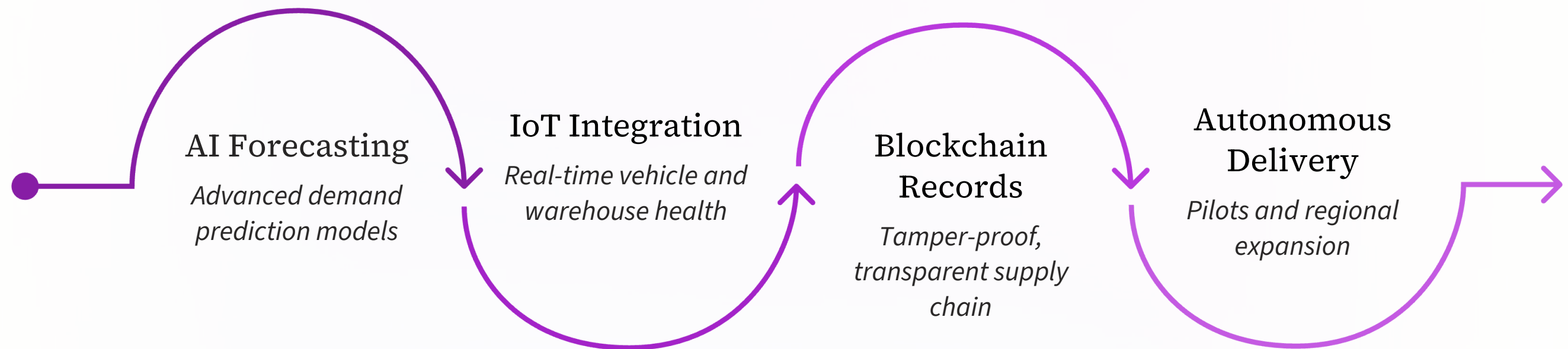


Scalable & Future-Ready

AI-driven insights optimize routes, resources, and financials. Modular design supports growth and adapts to seasonal demand.



Future Roadmap



The platform's evolution follows a clear trajectory — from intelligent forecasting and IoT connectivity to blockchain-backed transparency and autonomous delivery capabilities, positioning the system for sustained growth and regional scale.

Key Takeaways

Role-Based Dashboards

Tailored interfaces for every stakeholder with real-time intelligence and granular access control.

Smart Fleet & Warehouse

Automated resource assignment, optimized routing, live inventory tracking, and smart fulfillment.

AI-Powered Support & Generative AI

Always-on customer support, predictive analytics, fraud detection, and voice-activated assistance.

Financial Automation

Digital invoicing, real-time payment tracking, and automated payroll with expense auditing.

Scalable Future Vision

Modular architecture supporting IoT, blockchain, autonomous delivery, and regional expansion.

Thank You!

Let's Transform Rural Logistics, Together.

The floor is now open for questions, technical deep-dives, and feedback on the Cargo-Core platform. Feel free to scan the QR codes from the previous slide to access our live demo, code repository, and full project documentation.



We are QuadCore-Devs (Team-003) — building the infrastructure that moves what matters.

